Question 1

a) Giving all distinct values in cartesian representation for $\left(1+i\sqrt{3}\right)^{-9}$

$$R = \sqrt{(1)^2 + \left(\sqrt{3}\right)^2}$$
$$= \sqrt{1+3}$$
$$= \sqrt{4}$$
$$= 2$$

$$\theta = \tan^{-1}\frac{\sqrt{3}}{1}$$
$$= \frac{\pi}{3}$$

$$\left(1+i\sqrt{3}\right)^{-9} = \left(2e^{i\frac{\pi}{3}}\right)^{-9}$$
$$= \left[2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)\right]^{-9}$$
$$= 2^{-9}\left(\cos\left(-9\times\frac{\pi}{3}\right) + i\sin\left(-9\times\frac{\pi}{3}\right)\right)$$
$$= \frac{1}{2^9}(\cos(-3\pi) + i\sin(-3\pi))$$
$$= \frac{1}{512}(\cos 3\pi - i\sin 3\pi)$$
$$= \frac{1}{512}(\cos\pi - i\sin\pi)$$
$$= \frac{1}{512}(-1-0i)$$
$$= -\frac{1}{512}$$

$$\therefore \quad \left(1+i\sqrt{3}\right)^{-9} = -\frac{1}{512}$$

b) Giving all distinct values in cartesian representation for $(4i)^{-\frac{1}{4}}$

$$R = 4$$

$$\theta = \frac{\pi}{2}$$

$$(4i)^{-\frac{1}{4}} = \left(4e^{i\left[\frac{\pi}{2}+2n\pi\right]}\right)^{-\frac{1}{4}}$$

$$= 4^{-\frac{1}{4}}\, e^{i\frac{\left[\frac{\pi}{2}+2n\pi\right]}{-4}}$$

$$= \frac{1}{4^{\frac{1}{4}}}\, e^{-i\frac{\pi}{8}}\, e^{-i\frac{n\pi}{2}}$$

$$= \frac{1}{(2^2)^{\frac{1}{4}}}\left(\cos\left(-\frac{\pi}{8}\right) + i\sin\left(-\frac{\pi}{8}\right)\right) e^{-i\frac{n\pi}{2}}$$

$$= \frac{1}{\sqrt{2}}\left(\cos\frac{\pi}{8} - i\sin\frac{\pi}{8}\right) e^{-i\frac{n\pi}{2}}$$

$$= \frac{\sqrt{2}}{2}\left(\cos\frac{\frac{\pi}{4}}{2} - i\sin\frac{\frac{\pi}{4}}{2}\right) e^{-i\frac{n\pi}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$$

$$= \frac{\sqrt{2}}{2}\left(\sqrt{\frac{1+\cos\frac{\pi}{4}}{2}} - i\sqrt{\frac{1-\cos\frac{\pi}{4}}{2}}\right) e^{-i\frac{n\pi}{2}}$$

$$= \frac{\sqrt{2}}{2}\left(\sqrt{\frac{1+\frac{\sqrt{2}}{2}}{2}} - i\sqrt{\frac{1-\frac{\sqrt{2}}{2}}{2}}\right) e^{-i\frac{n\pi}{2}}$$

$$= \frac{\sqrt{2}}{2}\left(\sqrt{\frac{\frac{2+\sqrt{2}}{2}}{2}} - i\sqrt{\frac{\frac{2-\sqrt{2}}{2}}{2}}\right) e^{-i\frac{n\pi}{2}}$$

$$= \frac{\sqrt{2}}{2}\left(\sqrt{\frac{2+\sqrt{2}}{4}} - i\sqrt{\frac{2-\sqrt{2}}{4}}\right) e^{-i\frac{n\pi}{2}}$$

$$= \frac{\sqrt{2}}{2}\left(\frac{\sqrt{2+\sqrt{2}}}{2} - i\frac{\sqrt{2-\sqrt{2}}}{2}\right) e^{-i\frac{n\pi}{2}}$$

$$= \left(\frac{\sqrt{2\left(2+\sqrt{2}\right)}}{4} - i\frac{\sqrt{2\left(2-\sqrt{2}\right)}}{4}\right) e^{-i\frac{n\pi}{2}}$$

$$(4i)^{-\frac{1}{4}} = \left(\frac{\sqrt{4+2\sqrt{2}}}{4} - i\frac{\sqrt{4-2\sqrt{2}}}{4}\right) e^{-i\frac{n\pi}{2}}, \quad n \in \mathbb{Z}$$

$when\ n = 0$

$$z_1 = \left(\frac{\sqrt{4+2\sqrt{2}}}{4} - i\frac{\sqrt{4-2\sqrt{2}}}{4}\right)e^{-i\frac{0\cdot\pi}{2}}$$
$$= \frac{\sqrt{4+2\sqrt{2}}}{4} - \frac{\sqrt{4-2\sqrt{2}}}{4}i$$

$when\ n = 1$

$$z_2 = \left(\frac{\sqrt{4+2\sqrt{2}}}{4} - i\frac{\sqrt{4-2\sqrt{2}}}{4}\right)e^{-i\frac{\pi}{2}}$$
$$= \left(\frac{\sqrt{4+2\sqrt{2}}}{4} - i\frac{\sqrt{4-2\sqrt{2}}}{4}\right) \times -i$$
$$= -\frac{\sqrt{4-2\sqrt{2}}}{4} - \frac{\sqrt{4+2\sqrt{2}}}{4}i$$

$when\ n = 2$

$$z_3 = \left(\frac{\sqrt{4+2\sqrt{2}}}{4} - i\frac{\sqrt{4-2\sqrt{2}}}{4}\right)e^{-i\frac{2\pi}{2}}$$
$$= \left(\frac{\sqrt{4+2\sqrt{2}}}{4} - i\frac{\sqrt{4-2\sqrt{2}}}{4}\right)e^{-i\pi}$$
$$= \left(\frac{\sqrt{4+2\sqrt{2}}}{4} - i\frac{\sqrt{4-2\sqrt{2}}}{4}\right) \times -1$$
$$= -\frac{\sqrt{4+2\sqrt{2}}}{4} + \frac{\sqrt{4-2\sqrt{2}}}{4}i$$

$when\ n = 3$

$$z_4 = \left(\frac{\sqrt{4+2\sqrt{2}}}{4} - i\frac{\sqrt{4-2\sqrt{2}}}{4}\right)e^{-i\frac{3\pi}{2}}$$
$$= \left(\frac{\sqrt{4+2\sqrt{2}}}{4} - i\frac{\sqrt{4-2\sqrt{2}}}{4}\right) \times i$$
$$= \frac{\sqrt{4-2\sqrt{2}}}{4} + \frac{\sqrt{4+2\sqrt{2}}}{4}i$$

$$\therefore \quad (4i)^{-\frac{1}{4}} = \frac{\sqrt{4+2\sqrt{2}}}{4} - \frac{\sqrt{4-2\sqrt{2}}}{4}i, \quad -\frac{\sqrt{4-2\sqrt{2}}}{4} - \frac{\sqrt{4+2\sqrt{2}}}{4}i,$$
$$-\frac{\sqrt{4+2\sqrt{2}}}{4} + \frac{\sqrt{4-2\sqrt{2}}}{4}i, \quad \frac{\sqrt{4-2\sqrt{2}}}{4} + \frac{\sqrt{4+2\sqrt{2}}}{4}i$$

c) Giving all distinct values in cartesian representation for z such that $\cosh\sqrt{z} = 1$

$$\cosh\sqrt{z} = 1$$

$$\frac{e^{\sqrt{z}} + e^{-\sqrt{z}}}{2} = 1$$

$$e^{\sqrt{z}} + e^{-\sqrt{z}} = 2$$

$$\frac{\left(e^{\sqrt{z}}\right)^2}{e^{\sqrt{z}}} + \frac{1}{e^{\sqrt{z}}} = 2$$

$$\left(e^{\sqrt{z}}\right)^2 + 1 = 2e^{\sqrt{z}}$$

$$\left(e^{\sqrt{z}}\right)^2 - 2e^{\sqrt{z}} + 1 = 0$$

$$a^2 - 2ab + b^2 = (a-b)^2$$

$$(e^{\sqrt{z}} - 1)^2 = 0$$

$$e^{\sqrt{z}} - 1 = 0$$

$$e^{\sqrt{z}} = 1$$

$$e^{\sqrt{z}} = e^{i(0+2n\pi)}$$

$$\sqrt{z} = 2n\pi i$$

$$z = -4n^2\pi^2, \quad n \in \mathbb{Z}$$

Alternatively,

$$\cosh\sqrt{z} = 1$$

$$\cosh z = \cos iz$$

$$\cos i\sqrt{z} = 1$$

$$i\sqrt{z} = 2n\pi$$

$$\sqrt{z} = 2n\pi i$$

$$z = -4n^2\pi^2, \quad n \in \mathbb{Z}$$

yields the same result.

d) Giving lower and upper bounds for $|z-3i|$ if $|z+4|<\varepsilon$

$$\big||z_1|-|z_2|\big| \leq |z-3i| \leq |z_1|+|z_2|$$

$$\big||z_1|-|z_2|\big| \leq |z+4-4-3i| \leq |z_1|+|z_2|$$

$$\big||z+4|-|-4-3i|\big| \leq |(z+4)+(-4-3i)| \leq |z+4|+|-4-3i|$$

$$\begin{aligned}|-4-3i| &= \sqrt{(-4)^2+(-3)^2} \\ &= \sqrt{16+9} \\ &= \sqrt{25} \\ &= 5\end{aligned}$$

$$|\varepsilon-5| < |z-3i| < \varepsilon+5$$

$$|\varepsilon-5| = \begin{cases} \varepsilon-5, & \varepsilon \geq 5 \\ -(\varepsilon-5), & \varepsilon < 5 \end{cases}$$

$$\therefore \quad \varepsilon-5 < |z-3i| < \varepsilon+5, \quad if\ \varepsilon \geq 5$$

$$5-\varepsilon < |z-3i| < \varepsilon+5, \quad if\ \varepsilon < 5$$

Question 2

a) Using the Cauchy-Riemann equations to show that $w(z) = z^{-1}$ is analytic in $\mathbb{C}\backslash\{0\}$ – the set of all complex numbers apart from 0

$$w(z) = \frac{1}{z}$$

$$= \frac{1}{x+iy}$$

$$= \frac{x-iy}{(x+iy)(x-iy)}$$

$$= \frac{x-iy}{x^2+y^2}$$

$$= \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}$$

$$w(z) = u + iv$$

$$\frac{\partial u}{\partial x} = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2}$$

$$= \frac{-x^2+y^2}{(x^2+y^2)^2}$$

$$\frac{\partial v}{\partial y} = \frac{-(x^2+y^2)+2y^2}{(x^2+y^2)^2}$$

$$= \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial u}{\partial y} = -\frac{2xy}{(x^2+y^2)^2}$$

$$\frac{\partial v}{\partial x} = \frac{2xy}{(x^2+y^2)^2}$$

Except at $x = y = 0$ (the origin), these functions u and v have well-defined and continuous partial derivatives satisfying the Cauchy-Riemann equations since

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad and \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Therefore, $w(z) = z^{-1}$ is analytic everywhere apart from 0.

$$w'(z) = \frac{du}{dx} + i\frac{dv}{dx}$$

$$w'(z) = \frac{y^2-x^2}{(x^2+y^2)^2} + i\frac{2xy}{(x^2+y^2)^2}$$

b) Showing that $f(z) = (\bar{z})^{-1}$ is not differentiable, both by using the Cauchy-Riemann equations and by a geometric argument.

$$
\begin{aligned}
f(z) &= \frac{1}{\bar{z}} \\
&= \frac{1}{x - iy} \\
&= \frac{x + iy}{(x - iy)(x + iy)} \\
&= \frac{x + iy}{x^2 + y^2} \\
&= \frac{x}{x^2 + y^2} + i\frac{y}{x^2 + y^2}
\end{aligned}
$$

$$f(z) = u + iv$$

Firstly, by the Cauchy-Riemann equations

$$\frac{\partial u}{\partial x} = \frac{-x^2 + y^2}{(x^2 + y^2)^2} \qquad \frac{\partial v}{\partial y} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial y} = -\frac{2xy}{(x^2 + y^2)^2} \qquad \frac{\partial v}{\partial x} = -\frac{2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y} \quad and \quad \frac{\partial u}{\partial y} \neq -\frac{\partial v}{\partial x}$$

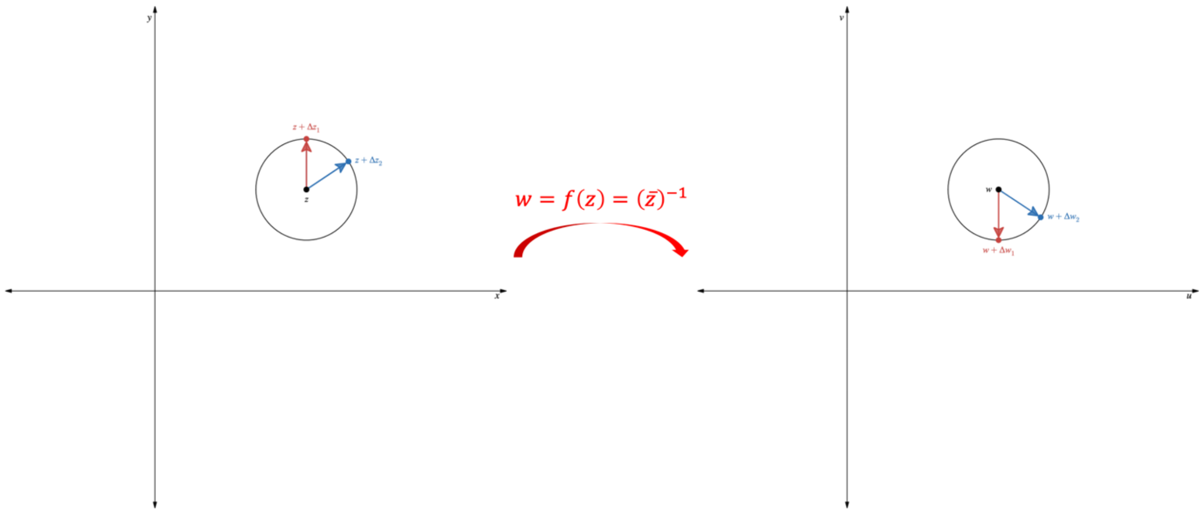
Therefore, $f(z) = (\bar{z})^{-1}$ is not differentiable since the Cauchy-Riemann equations don't hold.

Now, by a geometric argument.

If w is differentiable, then

$$w(z+\Delta z) \approx w(z) + \frac{dw(z)}{dz}\Delta z$$

$$\Longrightarrow \Delta w \approx \frac{dw(z)}{dz}\Delta z \quad \mathit{for\ some\ complex\ number}\ \frac{dw(z)}{dz}$$



In the above figure, $f(z) = (\bar{z})^{-1}$ maps the black point z, red point $z + \Delta z_1$ and blue point $z + \Delta z_2$ in the left-hand z-plane to the corresponding points in the right-hand w-plane. If $f'(z)$ exists, then rotating Δz by θ rotates Δw by θ. However in this case, to move from the red to the blue point in the z-plane, Δz is rotated $\frac{\pi}{4}$ clockwise; but in the w-plane, Δw is rotated $\frac{\pi}{4}$ anticlockwise. This switch in direction means there is no complex number $\frac{dw}{dz}$ such that $\Delta w = \frac{dw}{dz}\Delta z$, so the function cannot be differentiable at that, or indeed any, point.

c) By considering the proof of the Cauchy-Goursat theorem, showing that the area enclosed by a positively-oriented simple closed contour C is proportional to

$$\oint_C \bar{z}\ dz$$

$$\oint_C \bar{z}\ dz = \int_a^b \bar{z}(t)\ z'(t)\ dt$$

$$= \int_a^b [x - iy][x' + iy']\ dt$$

$$= \int_a^b [xx' + yy'] + i[-yx' + xy']\ dt$$

$$= \oint_C\ [xdx + ydy] + i \oint_C\ [-ydx + xdy]$$

$$\oint_C\ L\ dx + M\ dy = \iint_D \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y}\right)\ dA$$

$$= \iint_C\ \left[\frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x)\right]\ dA + i \iint_C\ \left[\frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(-y)\right]\ dA$$

$$= \iint_C\ [0 - 0]\ dA + i \iint_C\ [1 - (-1)]\ dA$$

$$= i \iint_C\ 2\ dA$$

$$\oint_C \bar{z}\ dz = 2i \times A_C$$

$$A_C = \frac{1}{2i} \oint_C \bar{z}\ dz$$

$$\therefore \quad A_C \propto \oint_C \bar{z}\ dz$$

$$\textit{where } \frac{1}{2i} \textit{ is the constant of proportionality}$$

Question 3

a) By parametrizing the contour path, evaluating

$$\oint_C z^n \, dz$$

where C is the positively-oriented unit circle where n is any integer

Using the parametrization of the form $z_0 + Re^{i\theta}$, for circle centre z_0, radius R and range of angles θ. The unit circle's centre is $z_0 = 0$, the radius is 1, and θ ranges from 0 to 2π as it completes one full revolution.

$Let\ \ z = e^{i\theta},\ \ 0 < \theta < 2\pi$

$With\ this\ parametrization,\ \ z^n = e^{in\theta}\ \ and\ \ z'(\theta) = ie^{i\theta}$

$$\begin{aligned}\oint_C z^n \, dz &= \int_0^{2\pi} z(\theta)^n \, \frac{dz(\theta)}{d\theta} \, d\theta \\ &= \int_0^{2\pi} e^{in\theta} \left(ie^{in\theta}\right) d\theta \\ &= i\int_0^{2\pi} e^{i(n+1)\theta} \, d\theta\end{aligned}$$

$when\ n \neq -1$

$$\begin{aligned}\oint_C z^n \, dz &= i\left[\frac{e^{i(n+1)\theta}}{i(n+1)}\right]_0^{2\pi} \\ &= \frac{e^{i(n+1)\cdot 2\pi} - e^{i(n+1)\cdot 0}}{n+1} \\ &= \frac{\cos(2\pi n + 2\pi) + i\sin(2\pi n + 2\pi) - 1}{n+1} \\ &= \frac{1 + i\sin(2\pi n) - 1}{n+1} \\ &= \frac{i\sin(2\pi n)}{n+1} \\ &= 0\ \ \ since\ n\ is\ any\ integer\end{aligned}$$

$when\ n = -1$

$$\begin{aligned}\oint_C z^n \, dz &= i\int_0^{2\pi} 1 \, d\theta \\ &= i[\theta]_0^{2\pi} \\ &= i[2\pi - 0] \\ &= 2\pi i\end{aligned}$$

$$\therefore\ \ \oint_C z^n \, dz = \begin{cases} 0 & n \in \mathbb{Z},\ \ n \neq -1 \\ 2\pi i, & n = -1 \end{cases}$$

b) Evaluating

$$\oint_C \frac{e^z\,dz}{z^2(z-3)}$$

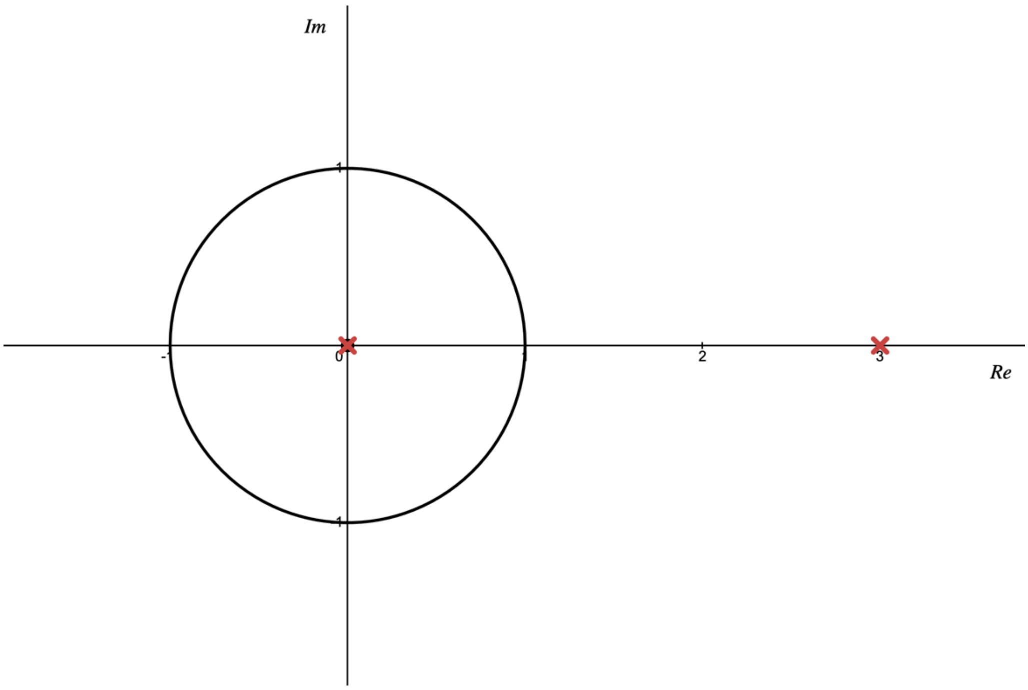
if C is the positively oriented unit circle

$\frac{e^z\,dz}{z^2(z-3)}$ has singularities where $z^2(z-3)=0$

$$z^2 = 0 \quad or \quad z-3=0$$

$$\therefore \quad at \quad z=0 \quad and \quad z=3$$

i. $|z| = 1$



Only the singularity at $z=0$ lies inside C. Therefore, using the generalized CIF since $z_0 = 0$ lies inside the contour C, and the function $f(z)$ is analytic on and everywhere inside C. The only point where $f(z)$ is not analytic is at $z=3$, which lies outside C.

$$f(z) = \frac{e^z}{z-3}$$

$$z_0 = 0$$

$$n = 1$$

$$\oint_C \frac{e^z\,dz}{z^2(z-3)} = \frac{2\pi i}{1!} f'(0)$$

$$f'(z) = \frac{e^z(z-3)^2 - e^z}{(z-3)^2}$$

$$f'(z) = \frac{e^z}{z-3} - \frac{e^z}{(z-3)^2}$$

$$\oint_C \frac{e^z\,dz}{z^2(z-3)} = 2\pi i\left[\frac{e^z}{z-3} - \frac{e^z}{(z-3)^2}\right]_{z=0}$$

$$= 2\pi i\left[\frac{e^0}{0-3} - \frac{e^0}{(0-3)^2}\right]$$

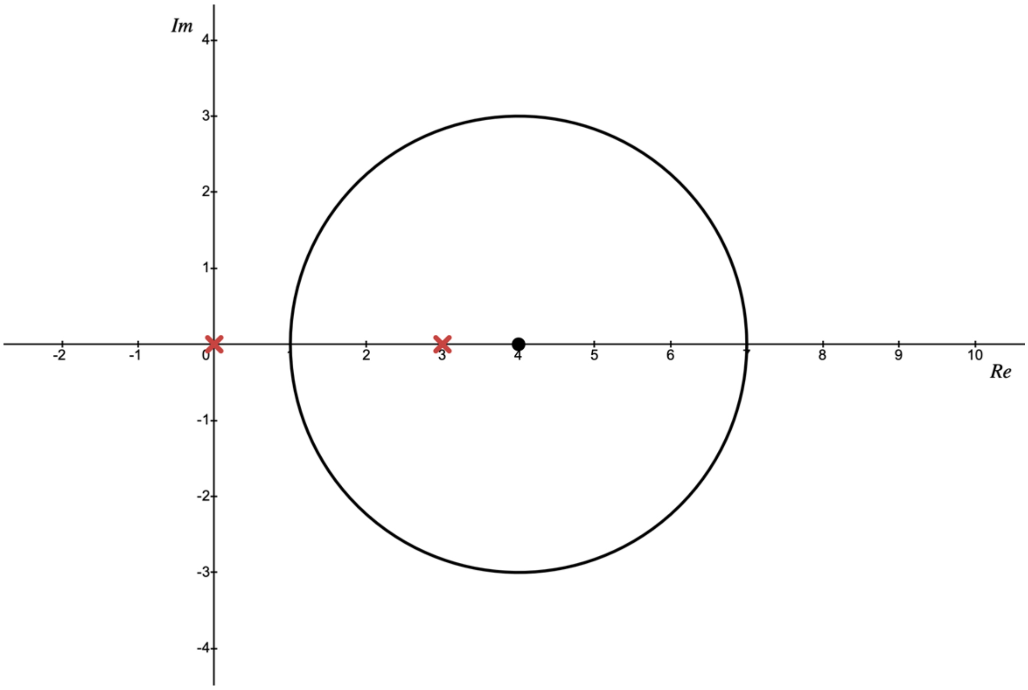
$$= 2\pi i\left(-\frac{1}{3} - \frac{1}{9}\right)$$

$$= 2\pi i\left(-\frac{3+1}{9}\right)$$

$$= 2\pi i\left(-\frac{4}{9}\right)$$

$$\oint_C \frac{e^z\,dz}{z^2(z-3)} = -\frac{8\pi}{9}i$$

ii. $|z-4|=3$



Only the singularity at $z=3$ lies inside C. Therefore, using the CIF since $z_0=3$ lies inside the contour C, and the function $f(z)$ is analytic on and everywhere inside C. The only point where $f(z)$ is not analytic is at $z=0$, which lies outside C.

$$f(z)=\frac{e^z}{z^2}$$

$$z_0=3$$

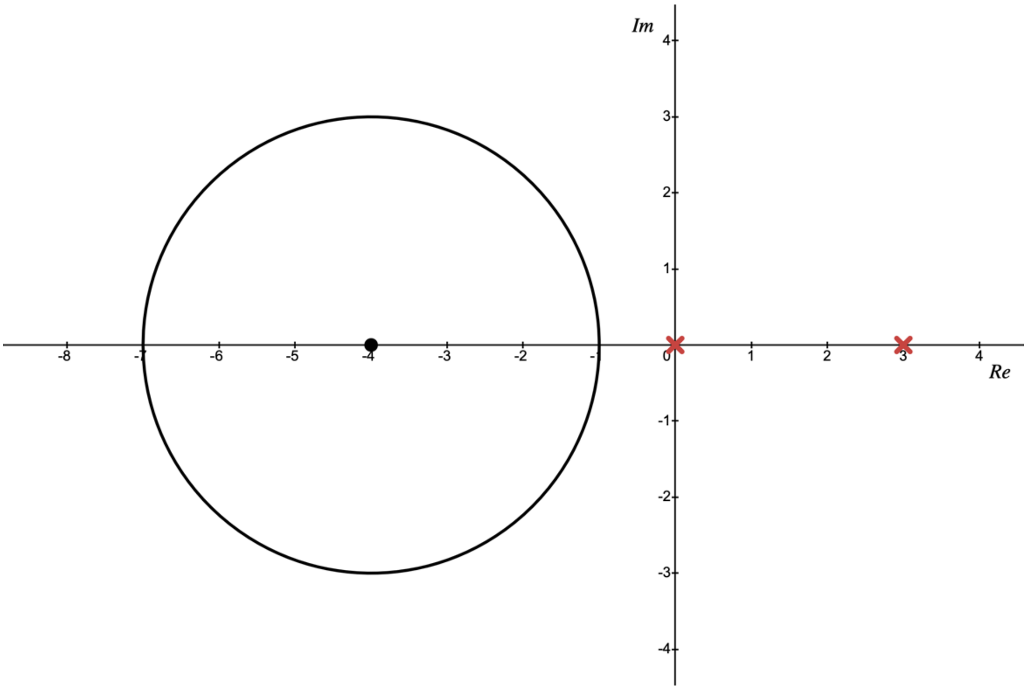
$$n=0$$

$$\oint_C \frac{e^z\,dz}{z^2(z-3)} = 2\pi i\, f(3)$$

$$= 2\pi i \times \frac{e^3}{3^2}$$

$$\oint_C \frac{e^z\,dz}{z^2(z-3)} = \frac{2\pi e^3}{9} i$$

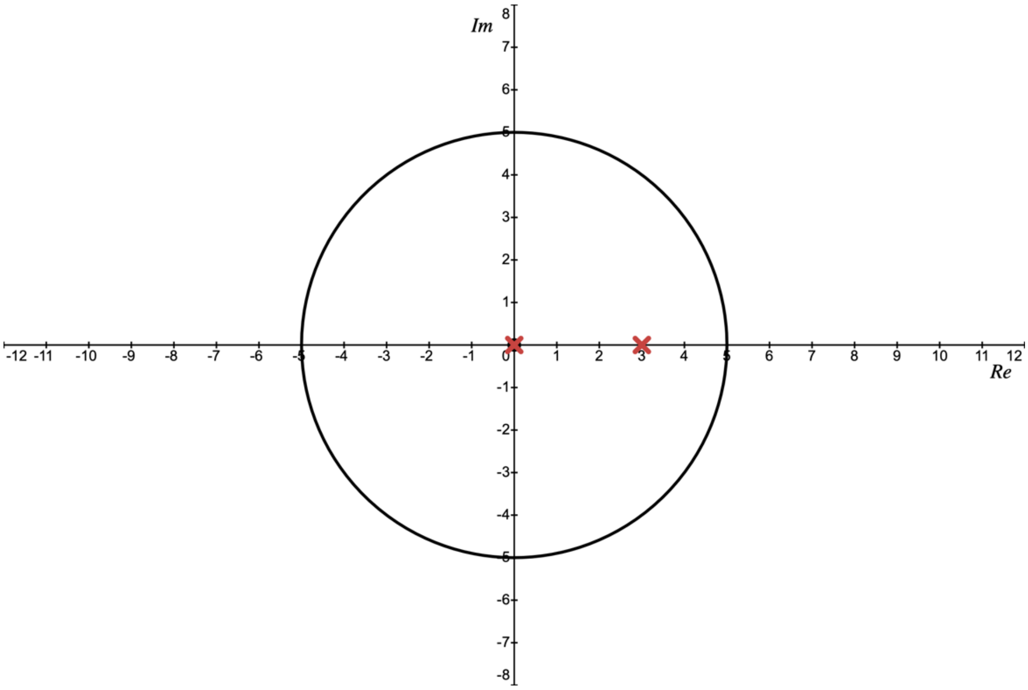
iii. $|z+4| = 3$



The singularities $z = 0$ and $z = 3$ both lie outside the contour C. Consequently, the numerator and $\frac{1}{z^2(z-3)}$ are both functions analytic on and in C. Thus, the whole integrand is analytic on and inside C, so by the Cauchy-Goursat theorem,

$$\oint_C \frac{e^z\, dz}{z^2(z-3)} = 0$$

iv. $|z| = 5$



The singularities $z = 0$ and $z = 3$ both lie inside the contour C.

$$\oint_C \frac{e^z\,dz}{z^2(z-3)} = 2\pi i \sum_{k=1}^{2} \mathrm{Res}_{z=z_1} \frac{e^z}{z^2(z-3)}$$

$$= 2\pi i \left[\mathrm{Res}_{z=0} \frac{\frac{e^z}{z-3}}{z^2} + \mathrm{Res}_{z=3} \frac{\frac{e^z}{z^2}}{z-3} \right]$$

$$= 2\pi i\ \mathrm{Res}_{z=0} \frac{\phi_i(z)}{z^2} + 2\pi i\ \mathrm{Res}_{z=3} \frac{\phi_{ii}(z)}{z-3} \quad (\textit{the } \phi \textit{ rule is just a version of the CIF})$$

$$= -\frac{8\pi}{9} i + \frac{2\pi e^3}{9} i \qquad (from\ part\ i.\ \&\ ii.\ respectively)$$

$$\oint_C \frac{e^z\,dz}{z^2(z-3)} = \frac{2\pi e^3 - 8\pi}{9} i$$

c) If C is an arbitrary positively-oriented closed contour, and

$$g(z) = \oint_C \left(\frac{s}{s-z}\right)^2 \, ds$$

finding what values $g(z)$ takes in the complex plane

$$g(z) = \oint_C \frac{s^2}{(s-z)^2} \, ds$$

For z inside C

The numerator is analytic everywhere in the complex plane, it is an entire function. Therefore, $f(s)$ is analytic on and everywhere inside the contour C. Thus, the generalised CIF can be used whenever z lies inside C.

$f(s) = s^2$

$s_0 = z$

$n = 1$

$$g(z) = \oint_C \frac{s^2}{(s-z)^2} \, ds = \frac{2\pi i}{1!} \frac{d}{dz}(z^2)$$

$$= 2\pi i \, (2z)$$

$$= 4\pi i z$$

For z outide C

When z lies outside C, the numerator and $\frac{1}{(s-z)^2}$ are both functions analytic on and in C. Therefore, the whole integrand is analytic on and inside C, so by the Cauchy-Goursat theorem,

$$g(z) = \oint_C \frac{s^2}{(s-z)^2} \, ds = 0$$

Question 4

Part A

i) Using complex integration techniques to evaluate

$$\int_0^{2\pi} \frac{d\theta}{(3+\cos\theta)^2}$$

To solve this integral, let $z = e^{i\theta}$ and from the definition of $\cos\theta$

$$\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2} = \frac{z+z^{-1}}{2}$$

so that

$$\begin{aligned}(3+\cos\theta)^2 &= \left(3+\frac{1}{2}\left[z+\frac{1}{z}\right]\right)^2 \\ &= \left(3+\frac{z}{2}+\frac{1}{2z}\right)^2 \\ &= \left(\frac{z^2+6z+1}{2z}\right)^2 \\ &= \frac{(z^2+6z+1)^2}{4z^2}\end{aligned}$$

Also, the integral of $0 < \theta < 2\pi$ will become an integral of z around the unit circle, which is denoted by C, and

$$\frac{dz}{d\theta} = ie^{i\theta} = iz \Longrightarrow d\theta = \frac{dz}{iz}$$

Changing the variable of integration from θ to z

$$\begin{aligned}\int_0^{2\pi} \frac{d\theta}{(3+\cos\theta)^2} &= \oint_C \frac{1}{\frac{(z^2+6z+1)^2}{4z^2}} \frac{dz}{iz} \\ &= \oint_C \frac{4z^2}{(z^2+6z+1)^2} \frac{dz}{iz} \\ &= -4i \oint_C \frac{z}{(z^2+6z+1)^2}\, dz\end{aligned}$$

To evaluate this integral, the singularities of the integrand need to be found, and apply the Cauchy residue theorem to the singularities that lie inside C (i.e. inside the unit circle). The singularities occur where the denominator is zero, i.e. where

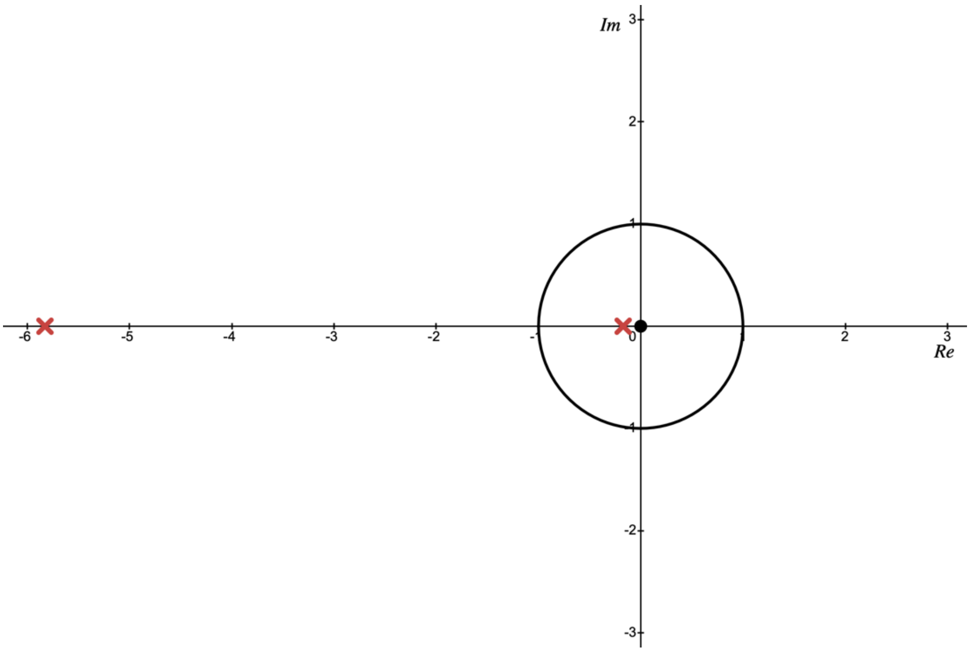
$$z^2 + 6z + 1 = 0$$

$$z = \frac{-(6) \pm \sqrt{(6)^2 - 4(1)(1)}}{2(1)}$$

$$= \frac{-6 \pm \sqrt{32}}{2}$$

$$= \frac{-6 \pm \sqrt{16} \cdot \sqrt{2}}{2}$$

$$z = -3 \pm 2\sqrt{2}$$



Both these solutions lie on the negative real axis. Therefore, only $-3 + 2\sqrt{2}$ is inside the unit circle since $-3 - 2\sqrt{2} < -1$. Denoting the relevant solution that is inside the unit circle z_1, yields

$$z_1 = -3 + 2\sqrt{2}\,, \qquad z_2 = -3 - 2\sqrt{2}$$

and

$$\int_0^{2\pi} \frac{d\theta}{(3 + \cos\theta)^2} = -4i \oint_C \frac{z}{(z^2 + 6z + 1)^2}\, dz$$

$$= -4i \times 2\pi i\ \operatorname{Res}_{z=z_1} \frac{z}{(z^2 + 6z + 1)^2}$$

However,

$$\frac{z}{(z^2+6z+1)^2} = \frac{z}{(z-z_1)^2(z-z_2)^2} = \frac{\frac{z}{(z-z_2)^2}}{(z-z_1)^2} = \frac{\phi(z)}{(z-z_1)^2}$$

Defining $\phi(z) = \frac{z}{(z-z_2)^2}$, then the ϕ-rule can be applied to evaluate the residue at z_1, where ϕ is analytic

$$\begin{aligned}\int_0^{2\pi} \frac{d\theta}{(3+\cos\theta)^2} &= 8\pi\ \operatorname{Res}_{z=z_1} \frac{\phi(z)}{(z-z_1)^2} \\ &= 8\pi \times \frac{\phi^{(2-1)}(z_1)}{(2-1)!} \\ &= 8\pi\ \phi'(z_1)\end{aligned}$$

Finding the first derivative of $\phi(z)$

$$\begin{aligned}\phi'(z) &= \frac{(z-z_2)^2 - 2z(z-z_2)}{(z-z_2)^4} \\ &= \frac{1}{(z-z_2)^2} - \frac{2z}{(z-z_2)^3} \\ &= \frac{z-z_2-2z}{(z-z_2)^3} \\ \phi'(z) &= \frac{-z-z_2}{(z-z_2)^3}\end{aligned}$$

Returning to the integral to be evaluated

$$\begin{aligned}\int_0^{2\pi} \frac{d\theta}{(3+\cos\theta)^2} &= 8\pi\ \left.\frac{-z-z_2}{(z-z_2)^3}\right|_{z=z_1} \\ &= 8\pi\ \frac{-z_1-z_2}{(z_1-z_2)^3} \\ &= 8\pi\ \frac{-(z_1+z_2)}{(z_1-z_2)^3} \\ &= 8\pi\ \frac{-\left(-3+2\sqrt{2}-3-2\sqrt{2}\right)}{\left(-3+2\sqrt{2}-\left[-3-2\sqrt{2}\right]\right)^3} \\ &= 8\pi\ \frac{-(-6)}{\left(-3+2\sqrt{2}+3+2\sqrt{2}\right)^3} \\ &= 8\pi\ \frac{6}{\left(4\sqrt{2}\right)^3}\end{aligned}$$

$$= \frac{48\pi}{64 \cdot 2\sqrt{2}}$$

$$= \frac{24\pi}{64\sqrt{2}}$$

$$= \frac{3\pi}{8\sqrt{2}}$$

$$= \frac{3\sqrt{2}\,\pi}{16}$$

$$\therefore \quad \int_0^{2\pi} \frac{d\theta}{(3+\cos\theta)^2} = \frac{3\sqrt{2}\,\pi}{16}$$

ii) Using complex integration techniques to evaluate the integral

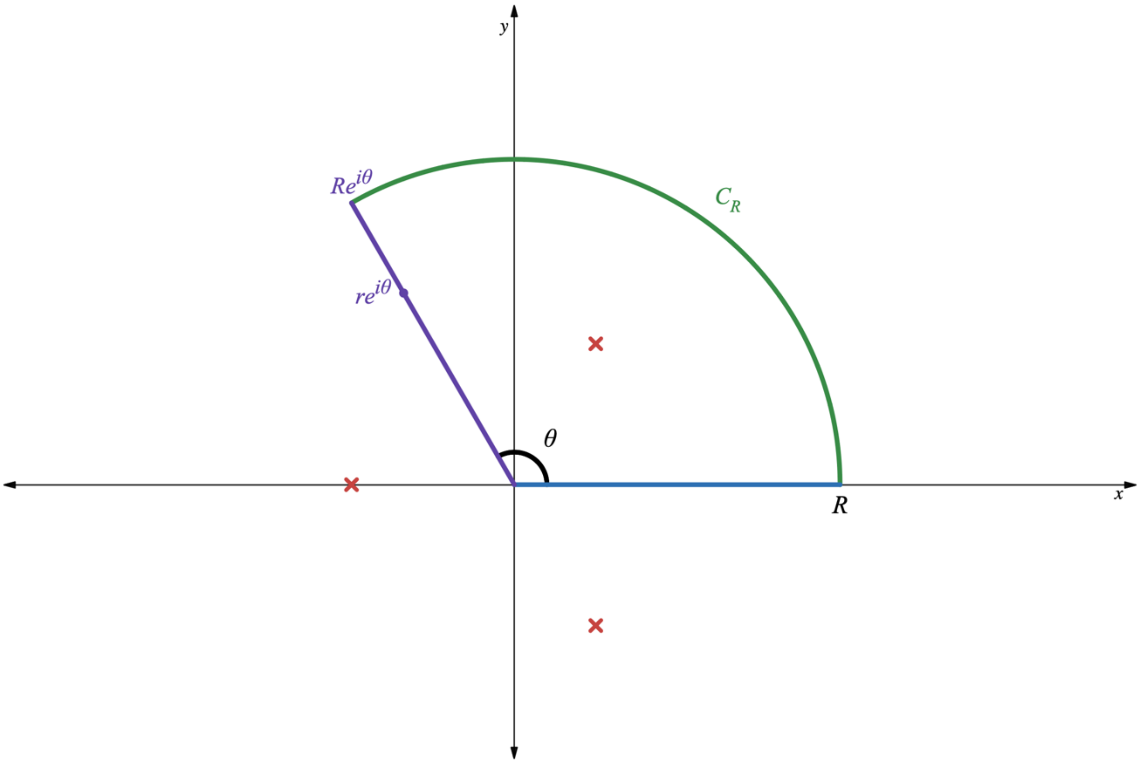
$$\int_0^\infty \frac{dx}{x^3 + 125}$$

To solve this problem, complex integration techniques will be applied to a contour in the form of a sector, whose angle $\theta = \frac{2\pi}{3}$.

$$Let\ I = \int_0^\infty \frac{dx}{x^3 + 125}$$

$$Let\ f(z) = \frac{1}{z^3 + 125}$$

The overall approach is to apply the CRT on the specifically-chosen sector-shaped contour as shown below.



On this contour,

$$\oint_C f(z)\,dz = \int_0^R f(x)\,dx + \int_{C_R} f(z)\,dz + \int_{Re^{i\frac{2\pi}{3}}}^0 f(z)\,dz = 2\pi i \sum_k \mathrm{Res}_{z=z_k} f(z)$$

Note that

$$\int_0^R f(x)\,dx = \int_0^R \frac{dx}{x^3 + 125} \xrightarrow{R\to\infty} I$$

Firstly, identifying the singularities of the integrand, which occur when

$$z^3 + 125 = 0$$

$$\begin{aligned} z^3 &= -125 \\ z_k &= \left(125e^{i[\pi+2k\pi]}\right)^{\frac{1}{3}} \\ &= 125^{\frac{1}{3}}\, e^{i\frac{[\pi+2k\pi]}{3}} \\ z_k &= 5e^{i\pi\left(\frac{1+2k}{3}\right)},\ \ k \in \mathbb{Z} \end{aligned}$$

$$z_1 = 5e^{i\pi\left(\frac{1+2(-1)}{3}\right)} = 5e^{-i\frac{\pi}{3}}$$

$$z_2 = 5e^{i\pi\left(\frac{1+2(0)}{3}\right)} = 5e^{i\frac{\pi}{3}}$$

$$z_3 = 5e^{i\pi\left(\frac{1+2(1)}{3}\right)} = 5e^{i\pi} = -5$$

$(choosing\ k = 0, \pm 1)$

The integrand only has one singularity in the upper half-plane that lies inside the contour as $R \to \infty$, which is the singularity at $z = 5e^{i\frac{\pi}{3}}$ (see the previous figure).

To find the residue of $f(z)$ at the relevant singularity, the p-over-q rule can be applied by defining the following

$$p(z) = 1 \neq 0$$

$$q(z) = z^3 + 125\big|_{z=5e^{i\frac{\pi}{3}}} = 0$$

$$q'(z) = 3z^2\big|_{z=5e^{i\frac{\pi}{3}}} \neq 0$$

These functions satisfy the p-over-q criteria. Therefore, it can be used to find the residue which is given by

$$\begin{aligned} \operatorname{Res}_{z=5e^{i\frac{\pi}{3}}} f(z) &= \frac{1}{3z^2}\bigg|_{z=5e^{i\frac{\pi}{3}}} \\ &= \frac{1}{3\left(5e^{i\frac{\pi}{3}}\right)^2} \\ &= \frac{1}{3\left(25e^{i\frac{2\pi}{3}}\right)} \\ &= \frac{1}{75e^{i\frac{2\pi}{3}}} \end{aligned}$$

To show that the integral along C_R goes to zero as $R \to \infty$, the formula for the bound on the integral is used. The length L of the contour is $\theta \times R$.

$$\left|\int_{C_R} f(z)\,dz\right| \leq ML = \max_{z\in C_R}|f(z)|\,\theta R$$

$$\left|\int_{C_R} \frac{dz}{z^3+125}\right| \leq \frac{1}{||z|^3-125|}\frac{2\pi}{3}R \leq \frac{2\pi R}{3R^3-375} \xrightarrow{R\to\infty} 0$$

$$since\ the\ denominator\ grows\ much\ faster\ than\ the\ numerator\ as\ R \to \infty$$

If the bound on the integral goes to zero, the only possibility is that the integral itself goes to zero. Thus,

$$\int_{C_R} \frac{dz}{z^3+125} \xrightarrow{R\to\infty} 0$$

The integral along the straight arm of argument $\theta = \frac{2\pi}{3}$ can be evaluated using the following parametrization

$$Let\ \ z = re^{i\frac{2\pi}{3}},\ \ 0 < r < R$$

$$dz = e^{i\frac{2\pi}{3}}\,dr$$

Therefore,

$$\int_{Re^{i\frac{2\pi}{3}}}^{0} f(z)\,dz = \int_R^0 \frac{e^{i\frac{2\pi}{3}}\,dr}{\left(re^{i\frac{2\pi}{3}}\right)^3+125}$$

$$= \int_R^0 \frac{e^{i\frac{2\pi}{3}}\,dr}{r^3e^{i2\pi}+125}$$

$$= -e^{i\frac{2\pi}{3}}\int_0^R \frac{dr}{r^3+125} \xrightarrow{R\to\infty} -e^{i\frac{2\pi}{3}}\,I$$

Putting all these components together to obtain

$$2\pi i \times \frac{1}{75e^{i\frac{2\pi}{3}}} = I + 0 - e^{i\frac{2\pi}{3}}\,I$$

$$I\left(1-e^{i\frac{2\pi}{3}}\right) = \frac{2\pi i}{75e^{i\frac{2\pi}{3}}}$$

$$I\left(e^{i\frac{2\pi}{3}}-e^{i\frac{4\pi}{3}}\right) = \frac{2\pi i}{75}$$

$$I = \frac{2\pi i}{75\left(e^{i\frac{2\pi}{3}} - e^{i\frac{4\pi}{3}}\right)}$$

$$= \frac{2\pi i}{75\left[\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3} - \left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)\right]}$$

$$= \frac{2\pi i}{75\left[-\frac{1}{2} + i\frac{\sqrt{3}}{2} - \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)\right]}$$

$$= \frac{2\pi i}{75\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} + \frac{1}{2} + i\frac{\sqrt{3}}{2}\right)}$$

$$= \frac{2\pi i}{75\sqrt{3}i}$$

$$= \frac{2\pi}{75\sqrt{3}}$$

$$I = \frac{2\sqrt{3}\,\pi}{225}$$

$$\therefore \quad \int_0^\infty \frac{dx}{x^3 + 125} = \frac{2\sqrt{3}\,\pi}{225}$$